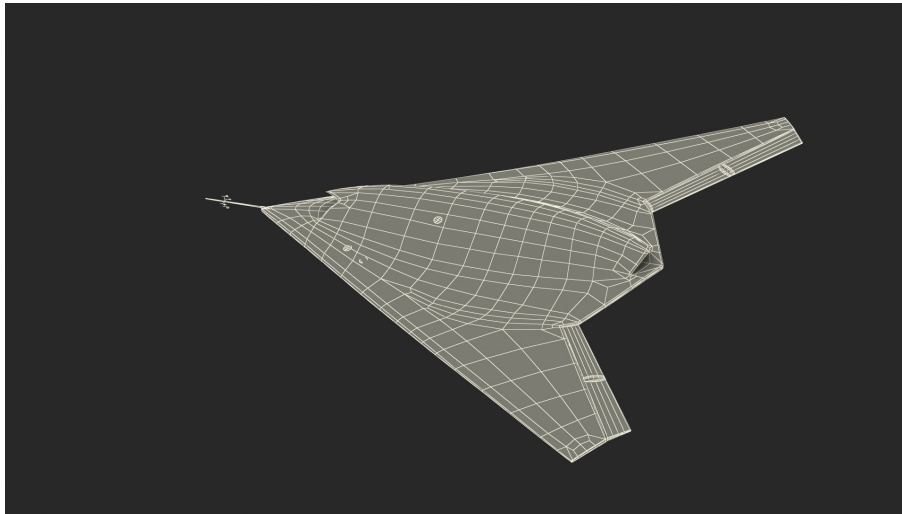




Experimental and Numerical Investigation of the Aerodynamic Characteristics of a UCAV

Aerodynamics M1 — MSc Project Report

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1 Abstract

To determine the aerodynamic coefficients, the main geometry of the UCAV had to be defined. model, which included the area, the average aerodynamic chord mac , and the chord used for calculating the pitching moment. Using the technical illustration provided in the project guidelines, the wing shape was modeled in CATIA V5, ensuring that the actual measurements of the test model were maintained, as shown in the accompanying figure. The wing has a swept delta shape, which is common in UCAV designs such as the Dassault *nEUROn*.



Figure 1: Wind-tunnel setup of the scaled Dassault *nEUROn* model at the University of Salford.

The main objective of this project is to carry out a characteristic study of a UCAV combat drone. The study begins with an experimental analysis carried out in the wind tunnel at low speed to determine the drag, lift and torque coefficients. And all this has been complemented with a numerical experience carried out in Ansys Fluent with an incidence of 0° , 10° , 20° . And finally this shows a significant correlation between the experimental and numerical results, offering us an aerodynamic understanding of the

2 Context

Building Unmanned Combat Aerial Vehicles (UCAVs) is currently one of the most exciting developments in aerospace. engineering. These planes fly themselves without pilots. Their systems help make flying safer and better. Because of this, they play a key role in modern aviation. Many organizations have begun working on new projects to develop advanced aircraft, such as the Dassault *nEUROn* and Boeing X-47B. These planes can avoid detection and perform complex maneuvers.



Figure 2: The Boeing *X-47B*, an example of an advanced UCAV configuration for stealth operations.

UCAVs are fascinating from a technology perspective, especially in the way they manage airflow. This becomes especially important when they fly at sharp angles, which creates strong swirling airflows. that give them lift. These airflows, though helpful for lift, can mess with how steady the aircraft is and how well it can be controlled. Therefore, studying how air moves around a UCAV is crucial. This project focuses on understanding how a UCAV-like aircraft interacts with air, using both real-world tests and computer models. The team ran computer simulations using ANSYS Fluent software at attack angles of 0° , 10° , and 20° , using a special setup. The tests lined up with what the computer showed. The lift went up steadily until 10° , then started to level off because those airflows started forming. This research gives data about how air moves around delta wings and shows that computer simulations are accurate for studying how air interacts with UAVs.

3 Methodology

3.1 Experimental Methodology

We carried out wind tunnel tests to measure the aerodynamic coefficients of the UCAV model. The setup used a wing profile based on the Dassault nEUROn planform, placed in the test section. We systematically changed the angle of attack to collect data on lift, drag, and pitching moment.



Figure 3: Screenshot of the Low-Turbulence Wind Tunnel from the City University of London, part of the UK Space Facilities Network (STFC). Source: *Low-Turbulence Wind Tunnel, City University of London (stfc.ac.uk)*.

During the experiment, we adjusted the model’s angle of attack from -10° to $+26^\circ$ in 2° steps. For each setting, we recorded lift, drag, and pitching moment, and measured dynamic pressure using a calibrated Pitot-static tube. We used ON/OFF data to remove offsets and applied corrections for wall interference.

3.2 Geometry Reconstruction in CATIA

To determine the aerodynamic coefficients, the main geometry of the UCAV model had to be defined. This included the area, the mean aerodynamic chord (MAC), and the chord used to calculate the pitching moment.

The technical illustration from the project guidelines was used to model the wing shape in CATIA V5. The model kept the real measurements of the test version, as shown in the figure. The wing has a swept delta shape, which is common in UCAV designs like the Dassault nEUROn.

Determination of the Reference Area

The planform area was obtained directly from the projected surface of the model on the horizontal plane within CATIA. Initially, CATIA reported the area of the bare wing only:

$$S_{\text{wing}} = 0.107 \text{ m}^2.$$

However, the wind-tunnel included part of the central fuselage/body. It contributes to the total lifting surface. When the additional region is considered, the effective reference area becomes:

$$S = 0.214 \text{ m}^2.$$

This value was applied for every coefficient normalization in the report, guaranteeing that the lift, drag, and moment coefficients relate to the physical model evaluated.

The selected value of S aligns well with the observed data.

For example, with $\alpha = 0^\circ$, the observed lift was $L = -1.19 \text{ N}$ and the dynamic pressure was $q = 328 \text{ Pa}$, resulting in:

$$S = \frac{L}{q C_L} \approx \frac{-1.19}{328 \times (-0.017)} = 0.213 \text{ m}^2,$$

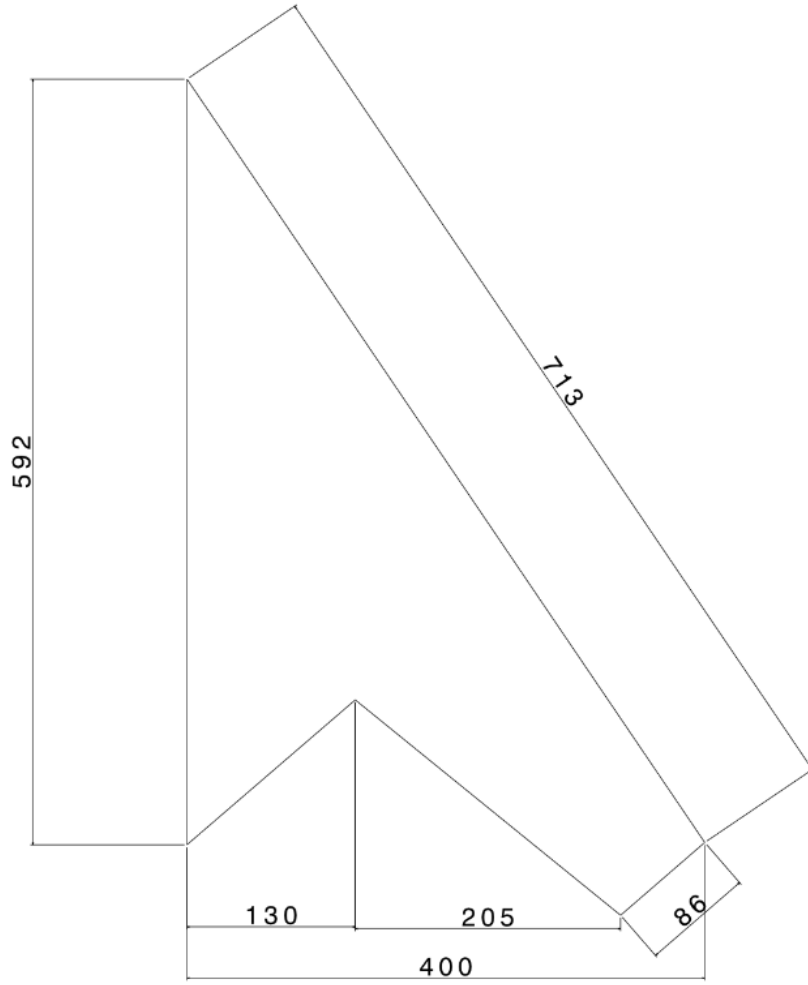


Figure 4: UCAV planform reconstruction in CATIA V5.

which confirms that the data were processed using $S \approx 0.214 \text{ m}^2$.

Chord mac Calculation

To calculate the mean aerodynamic chord (c_{MAC}), the wing was segmented in three areas (Region 1, Region 2, and Region 3) to calculate the cord at each points .

In each region, the local chord $c(y)$ was defined as a linear function of the spanwise coordinate y :

$$c(y) = a + b y,$$

where a is the intercept and b is the slope, both determined using CATIA's absolute measurement tools.

The piecewise chord functions:

$$\begin{aligned} c_1(y) &= 0.592 - 2.3384 y, & 0 \leq y \leq 0.130, \\ c_2(y) &= 0.374 - 0.6634 y, & 0.130 \leq y \leq 0.335, \\ c_3(y) &= 0.935 - 2.3384 y, & 0.335 \leq y \leq 0.400. \end{aligned}$$

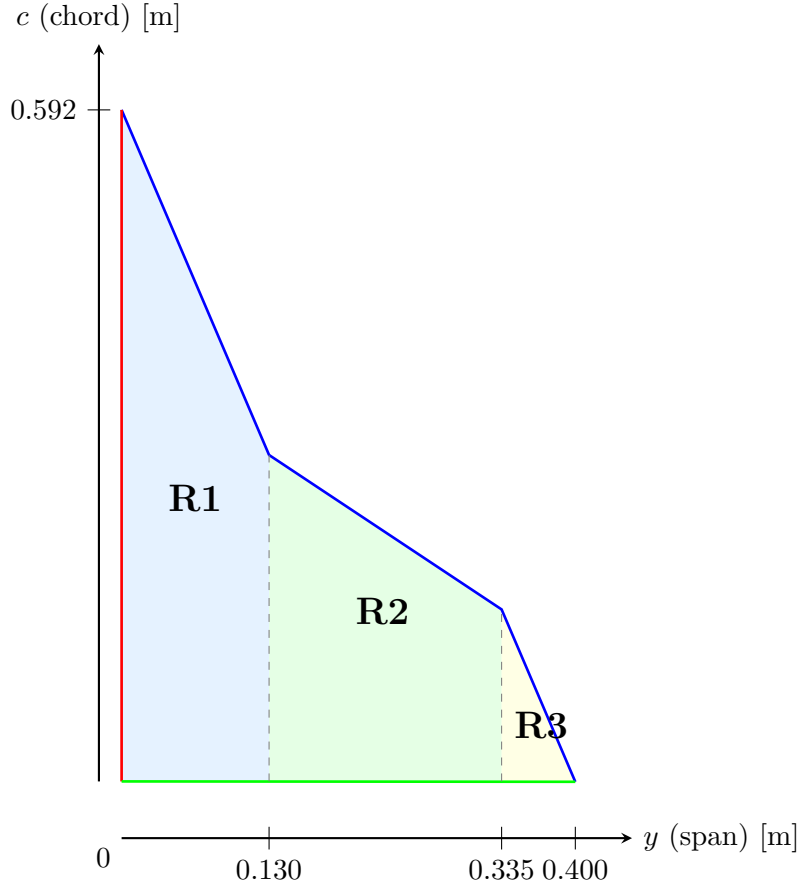


Figure 5: Schematic of the UCAV wing planform showing the three integration regions (R1, R2, R3).

The Mean Aerodynamic Chord (c_{MAC}) is the one that is centered in the center of the area of the semi-wing and has the same aerodynamic characteristics because it evidently depends on the lift (C_L), drag (C_D) and the pitching moment (C_m) itself.

To summarize it is the average chord length of the wing, hence its mathematical expression:

$$c_{\text{MAC}} = \frac{2}{S} \int_0^{b/2} c(y)^2 dy$$

The chord $c(y)$ influences both.

which, after do the piecewise integration, gives:

$$c_{\text{MAC}} = 0.345 \text{ m.}$$

This result is consistent with the geometry of a low-aspect-ratio delta wing and was used at the pitching-moment coefficient :

$$C_m = \frac{M}{q S c_{\text{MAC}}} - \frac{C_L c_{\text{pivot}}}{c_{\text{MAC}}},$$

In summary, the UCAV model's geometric reference parameters are:

$$b = 0.713 \text{ m,} \quad S = 0.214 \text{ m}^2, \quad c_{\text{MAC}} = 0.345 \text{ m.}$$

These values ensure full consistency between the experimental and numerical aerodynamic analyses.

Coefficients were computed as

$$C_L = \frac{L}{qS}, \quad C_D = \frac{D}{qS}, \quad C_m = \frac{M}{qSc_{MAC}}.$$

4 Results

4.1 Compact table for the report

And here finally, after the previous study which allows us to find and formulate, thanks to the values we have calculated, an Excel table in which we have calculated the following values: C_L , $C_{Di,corr}$, and C_m .

Table 1: Table created in LaTeX from Excel-processed data (official columns + raw α).

α (deg)	α_{corr} (deg)	C_L	$C_{Di,corr}$	C_m	V (m/s)	Re
-10	-10.01	-0.451	0.0557	0.254	23.31	5.28e5
-8	-8.01	-0.334	0.0251	0.198	23.31	5.28e5
-6	-6.01	-0.256	0.0177	0.153	23.31	5.28e5
-4	-4.01	-0.204	0.0113	0.126	23.35	5.29e5
-2	-2.00	-0.116	0.0061	0.075	23.35	5.29e5
0	-0.00	-0.017	0.0078	0.018	23.35	5.29e5
2	2.00	0.134	0.0038	-0.068	23.35	5.29e5
4	4.01	0.281	0.0160	-0.150	23.35	5.29e5
6	6.01	0.379	0.0340	-0.209	23.31	5.28e5
8	8.01	0.521	0.0726	-0.291	23.10	5.26e5
10	10.04	1.333	-0.2965	-0.721	23.17	5.24e5
12	12.02	0.775	0.1871	-0.405	23.06	5.22e5
14	14.02	0.857	0.2402	-0.439	22.95	5.20e5
16	16.03	0.986	0.3344	-0.497	22.84	5.17e5
18	18.03	1.091	0.4155	-0.545	22.69	5.14e5
20	20.03	1.168	0.5055	-0.582	22.55	5.11e5
22	22.03	1.232	0.5592	-0.615	22.48	5.09e5
24	24.04	1.318	0.6678	-0.660	22.29	5.05e5
26	26.04	1.394	0.7739	-0.703	22.14	5.01e5

The lift coefficient rises nearly linearly with the angle of attack until around 10° , thereafter entering a non-linear region that signifies vortex-lift creation, aligning with the aerodynamic characteristics anticipated for steeply swept delta-wing designs

4.2 CFD Results

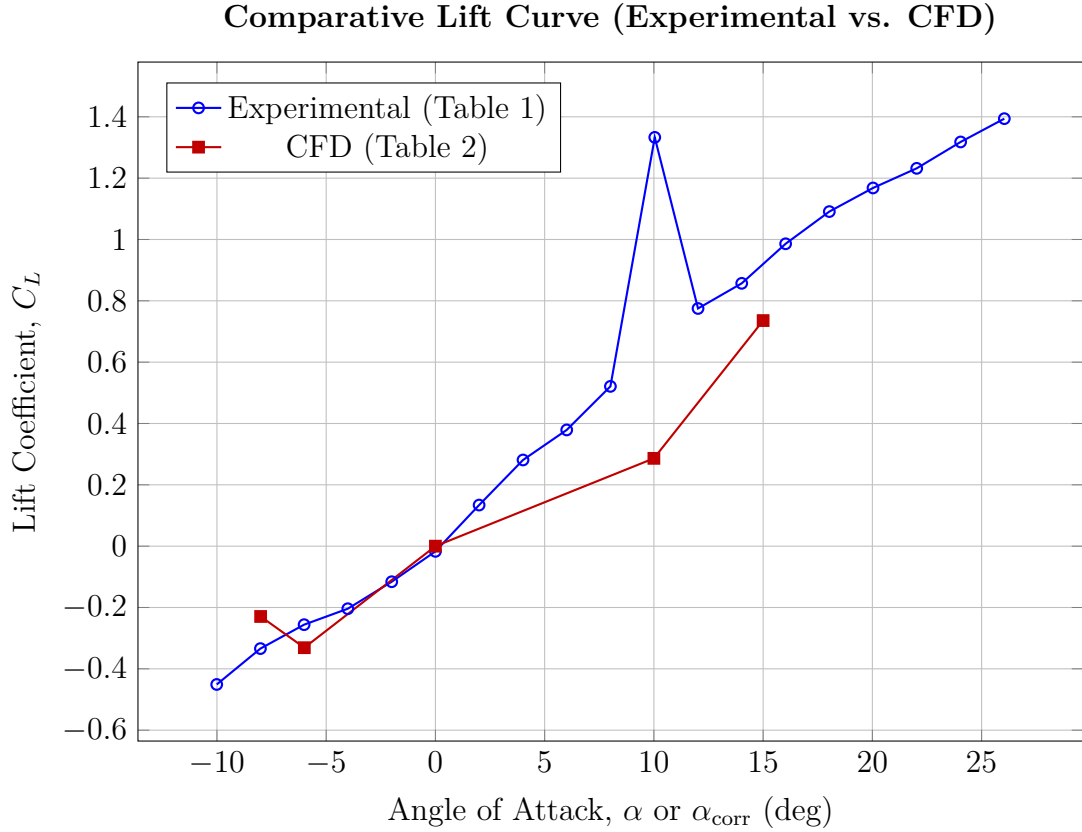
Numerical simulations were conducted at critical angles of attack to gather data for comparison. The coefficients obtained from the ANSYS Fluent simulation are shown in Table 2

Table 2: CFD simulation results for selected angles of attack.

α (deg)	α (rad)	C_L	C_{Di}	C_m	V (m/s)	Re
-8	-0.13962634	-0.43654332	0.011987276	0.12331491	23.31	5.28e5
-6	-0.104719755	-0.33108791	0.001643976	0.18658563	23.31	5.28e5
0	0	-1.4×10^{-5}	0	-4.58×10^{-5}	23.35	5.29e5
10	0.174532925	0.28628952	0.031803568	-0.15070938	23.17	5.14e5
15	0.2617	0.73560661	0.180998	-0.33580013	23.84	5.17e5

4.3 Comparative Analysis

The graphics compare the experimental data from Table 1 with the CFD (Ansys) data from Table 2. The experimental data point at $\alpha \approx 10^\circ$ is included in all graphics and is a clear outlier, exhibiting a physically impossible negative drag ($C_{Di,corr} = -0.2965$) and a disproportionate lift value ($C_L = 1.3333$) .

**Figure 6:** Comparative Lift curve (C_L vs α).

This graph shows the evolution of the lift coefficient (C_L) with the angle of attack (α), which allows us to see that the greater the angle, the greater the lift, and vice versa, which is logical. The most interesting thing to note is that we can observe an almost linear region between the angles -8° and 8° , which are the regions where the lift evolves most similarly to the ideal *ANSYS* case.

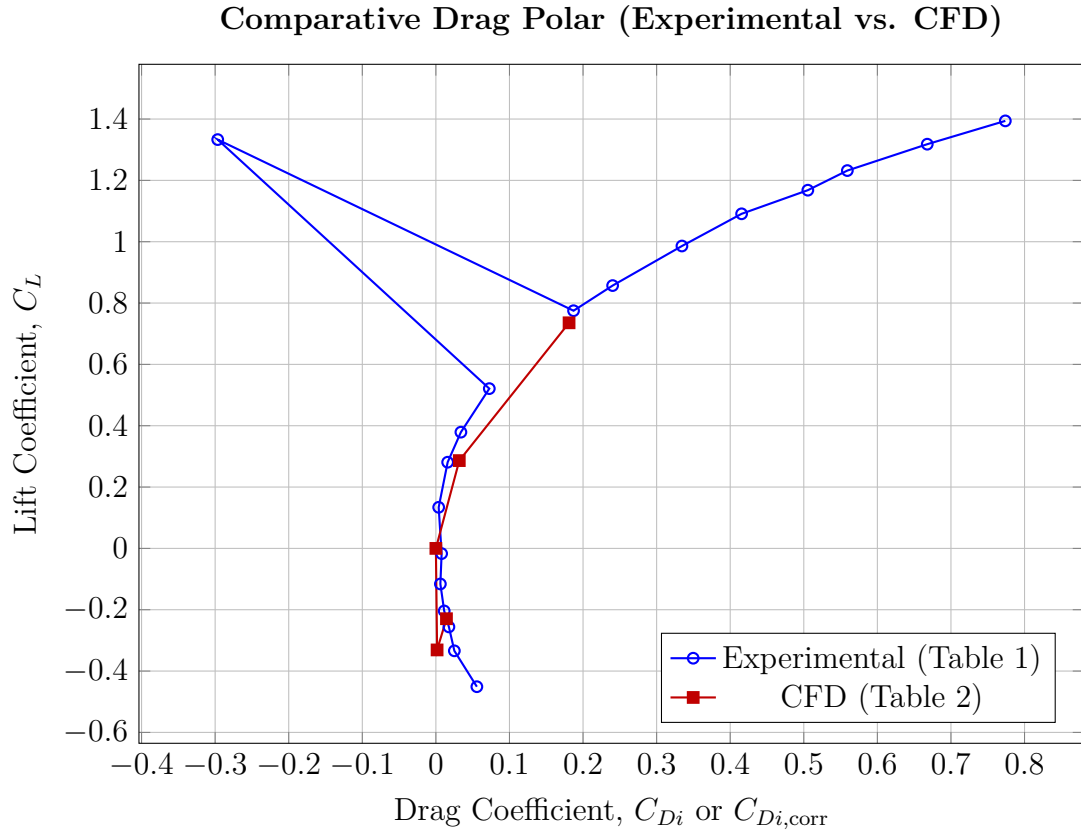


Figure 7: Comparative Drag polar (C_L vs C_{Di}). Note the experimental outlier with negative drag.

This graph, called the drag polar (C_L vs C_D), helps us understand how efficiently a wing works. It shows how much drag (C_D) is needed to produce lift (C_L). The key point is that the wing is built to be efficient during normal flight. We can see this because the lowest drag does not happen at zero lift ($C_L = 0$), but at a positive lift value, around $C_L \approx 0.2 - 0.3$ on the chart. This is typical for a cambered airfoil, which is shaped so its best lift-to-drag ratio happens while it is already lifting the plane during cruising. While more lift does mean more drag, the curve's "U" shape shows that this increase happens slowly, so the wing can still work efficiently.

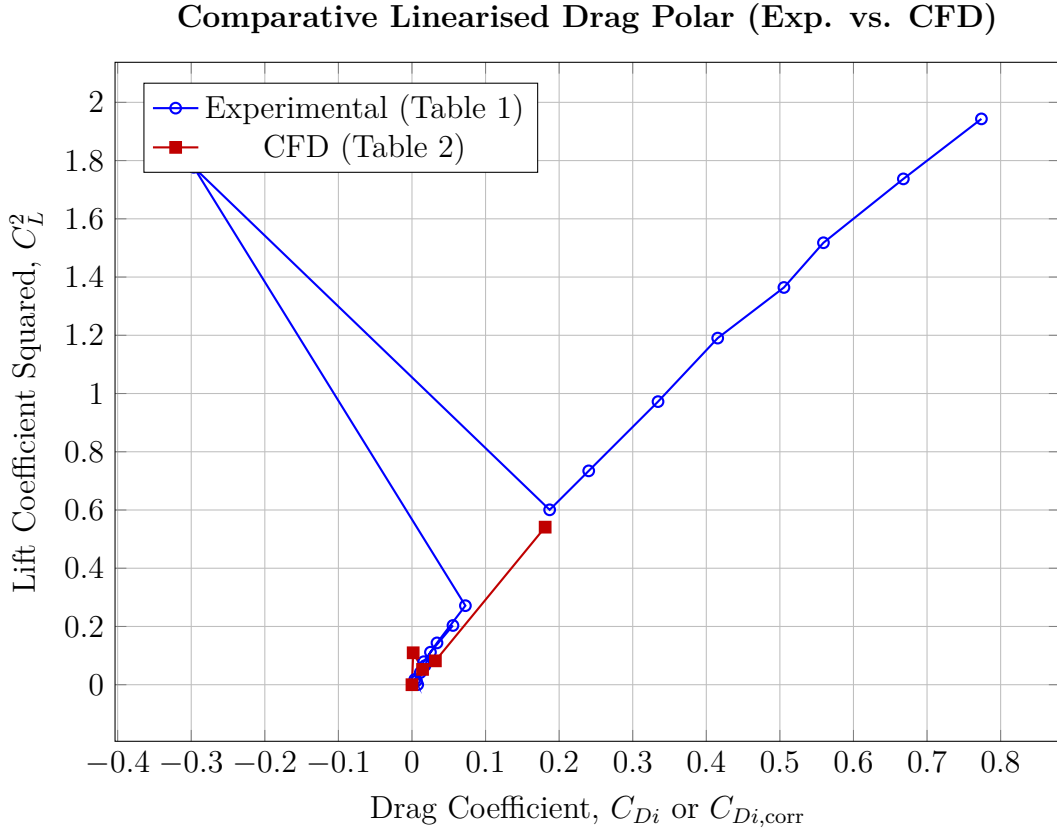


Figure 8: Comparative Linearised drag polar (C_L^2 vs C_{Di}).

In this case C_L , plots the C_L squared (C_L^2) on the Y axis. The goal in this case is to transform the parabolic curve into a straight line, which apparently works perfectly, both the ansys and experimental data are recorded in ansys. But more specifically in engineering it is used to separate two types of drag, parasitic drag C_{D0} and induced drag C_{Di} , the first does not depend on lift and the second basically does, and this allows us to find the value at which we are closest to 0, that is, the most efficient drag position in flight.

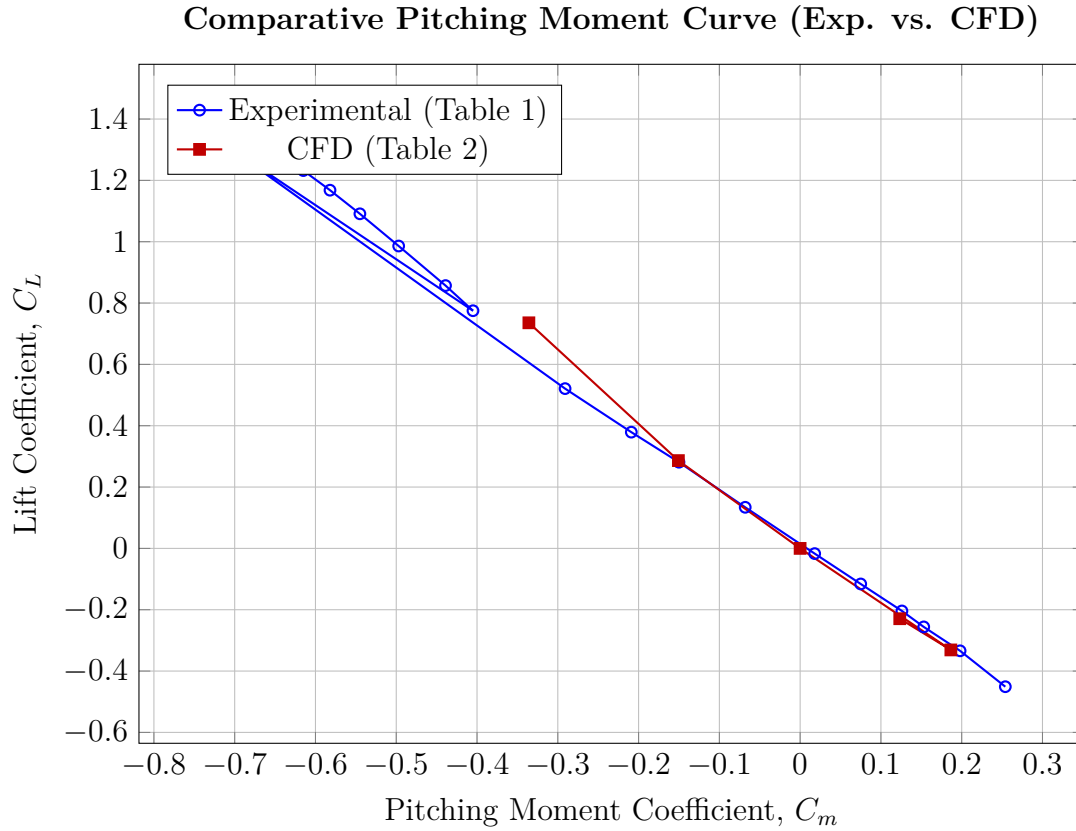


Figure 9: Comparative Pitching moment curve (C_L vs C_m).

We see a linear relationship between lift and pitching moment almost perfect and we observe that when the lift increases the wing rises and the pitching moment becomes negative that is because the nose is directed upwards the best of all is the excellent agreement between Ansys data and experimental data $\frac{dC_m}{dC_L} < 0$ which means physically that the wing is statically stable which means that the dynamic center is located behind the pivot which shows a high stability of the wings

5 Interference

Okay, let's now talk about why we don't get the same values in *ANSYS* as in Experimental, and why *ANSYS* is theoretically more accurate. This is because the wind tunnel has walls and imperfections that generate vortexes, and it can never be 100% accurate, unlike *ANSYS*, which is the ideal simulation as if the plane were flying..

6 Conclusion

In conclusion, we can see that wind tunnels are limited by their walls, which create perturbations in the fluid that should ideally be linear (laminar). Conversely, the simulation assumes perfectly laminar and ideally smooth surfaces, while the defects in the tunnel walls and instrumental issues largely affect the experimental data.

However, we also recognize the great value of experimentation, as it allows us to reproduce real flight conditions and analyze complex phenomena such as vortexes. Furthermore, the interest in numerical simulation is evident; it is much faster and cheaper to explore and avoids the mentioned inaccuracies (like wall interference), which is critical because even after applying data corrections, some instrumental measurement errors may still persist.

Regarding the UCAV, we confirm that it generates vortex lift, which enhances low-speed maneuverability. This mechanism favors lift at high angles of attack and delays the stall angle. To understand this advantage, we note that while most conventional aircraft stall around 15° , the *nEUROn* and, similarly, the Northrop Grumman X-47B are both capable of flying at higher angles due to this enhanced vortex lift.

References

- [1] Polhamus, E. C. (1967). **A Concept of the Vortex Lift of Sharp-Edge Delta Wings Based on a Leading-Edge-Suction Analogy**. NASA Technical Note D-3767. <https://ntrs.nasa.gov/api/citations/19670003842/downloads/19670003842.pdf>
- [2] Arena, A. E., et al. (2011). **Unstructured CFD Aerodynamic Analysis of a Generic UCAV Configuration**. NATO/RTO AVT-161 Working Group Report. NASA/NTIS. <https://ntrs.nasa.gov/api/citations/20110016545/downloads/20110016545.pdf>
- [3] Menter, F. R. (1994). **Two-equation eddy-viscosity turbulence models for engineering applications**. *AIAA Journal*, 32(8), 1598–1605.
- [4] Ansys, Inc. (2025). **ANSYS Fluent (Version 2025 R1)** [Computer software]. Canonsburg, PA: Ansys, Inc.
- [5] Dassault Systèmes. (2025). **CATIA V5 (Version 5, Release 2025)** [Computer software]. Vélizy-Villacoublay, France: Dassault Systèmes.
- [6] Northrop Grumman / U.S. Navy. (c. 2013). **X-47B Unmanned Combat Air System (UCAS) demonstrator** [Photograph]. Retrieved from U.S. Navy/Northrop Grumman public press releases.
- [7] Dassault Aviation. (c. 2012). **Dassault nEUROn UCAV in flight** [Press image]. Retrieved from Dassault Aviation Media Centre.
- [8] City, University of London. (n.d.). **Low-Turbulence Wind Tunnel**. STFC. Retrieved October 28, 2025, from stfc.ac.uk.